



US012414474B2

(12) **United States Patent**
Wise et al.

(10) **Patent No.:** **US 12,414,474 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **METHODS OF FORMING PIEZOELECTRIC MATERIALS, PIEZOELECTRIC DEVICES, AND ASSOCIATED TOOLING AND SYSTEMS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 754 days.

(21) Appl. No.: **17/507,279**

(22) Filed: **Oct. 21, 2021**

(65) **Prior Publication Data**
US 2023/0129396 A1 Apr. 27, 2023

(51) **Int. Cl.**
H10N 30/853 (2023.01)
H10N 30/078 (2023.01)

(52) **U.S. Cl.**
CPC **H10N 30/8554** (2023.02); **H10N 30/078** (2023.02)

(58) **Field of Classification Search**
CPC H10N 30/8554; H10N 30/078; H10N 30/092; H10N 30/081; H05K 3/125; H05K 1/09; B06B 1/0633; Y10T 29/42; Y10T 29/49155; B41C 1/144
See application file for complete search history.

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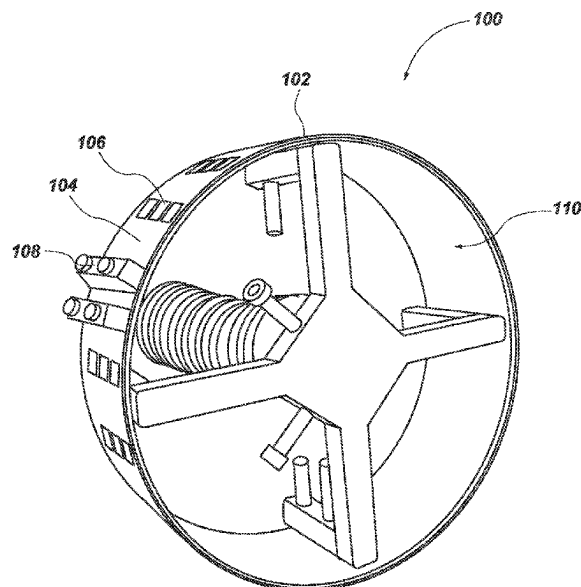
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(57) **ABSTRACT**

A method of forming a piezoelectric device may include depositing a sol-gel film over a substrate and curing the sol-gel film by impinging light onto an exposed surface of the sol-gel film to form a piezoelectric ceramic element. The method may produce a piezoelectric composite material including at least two piezoelectric ceramic pillars over the substrate. The at least two piezoelectric pillars may include at least one layer. The at least one layer having a gradient density, such that a first portion of the at least one layer proximate the substrate has a density lower than a second portion that is located a greater distance from the substrate than the first portion. The piezoelectric composite material may further include a resin separating the at least two piezoelectric ceramic pillars.

16 Claims, 11 Drawing Sheets



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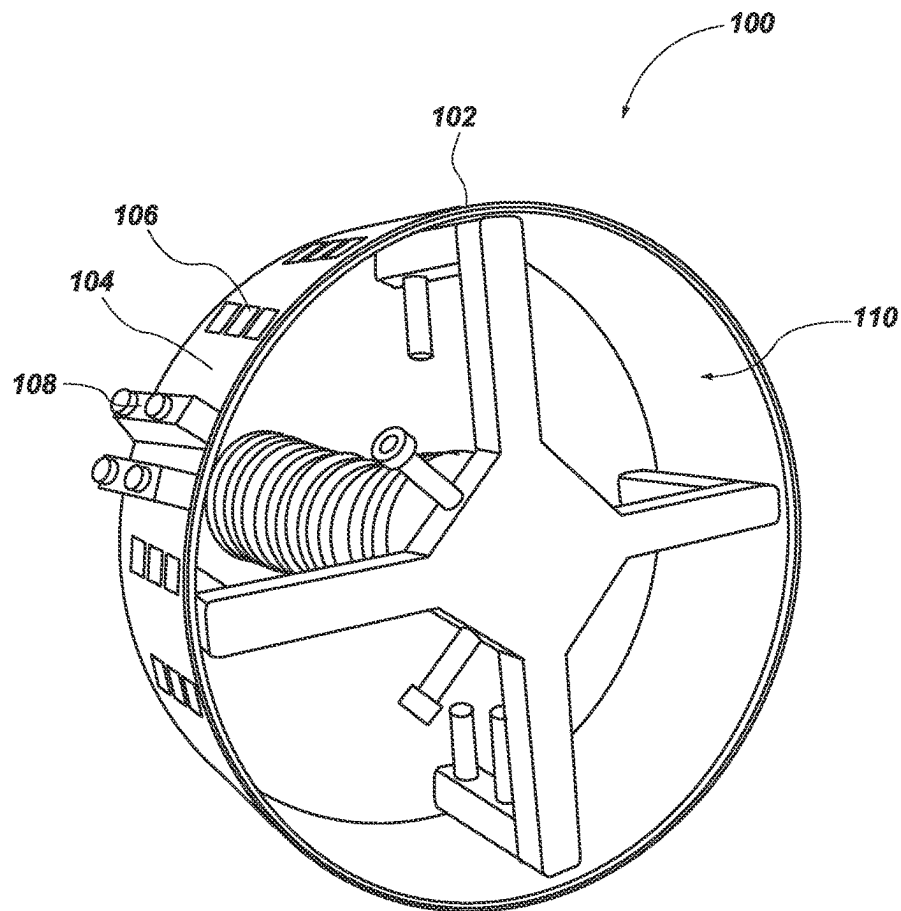
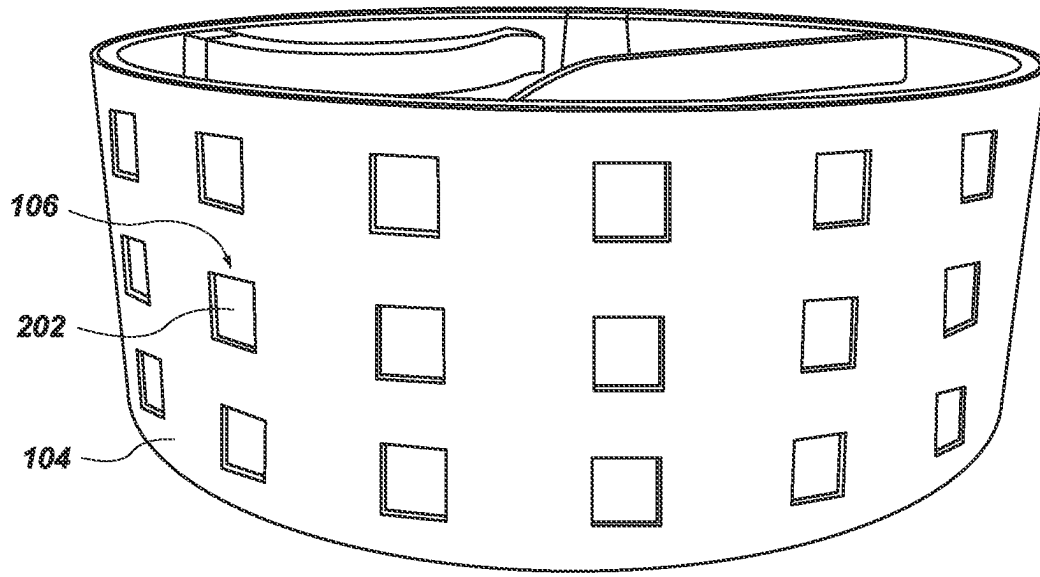
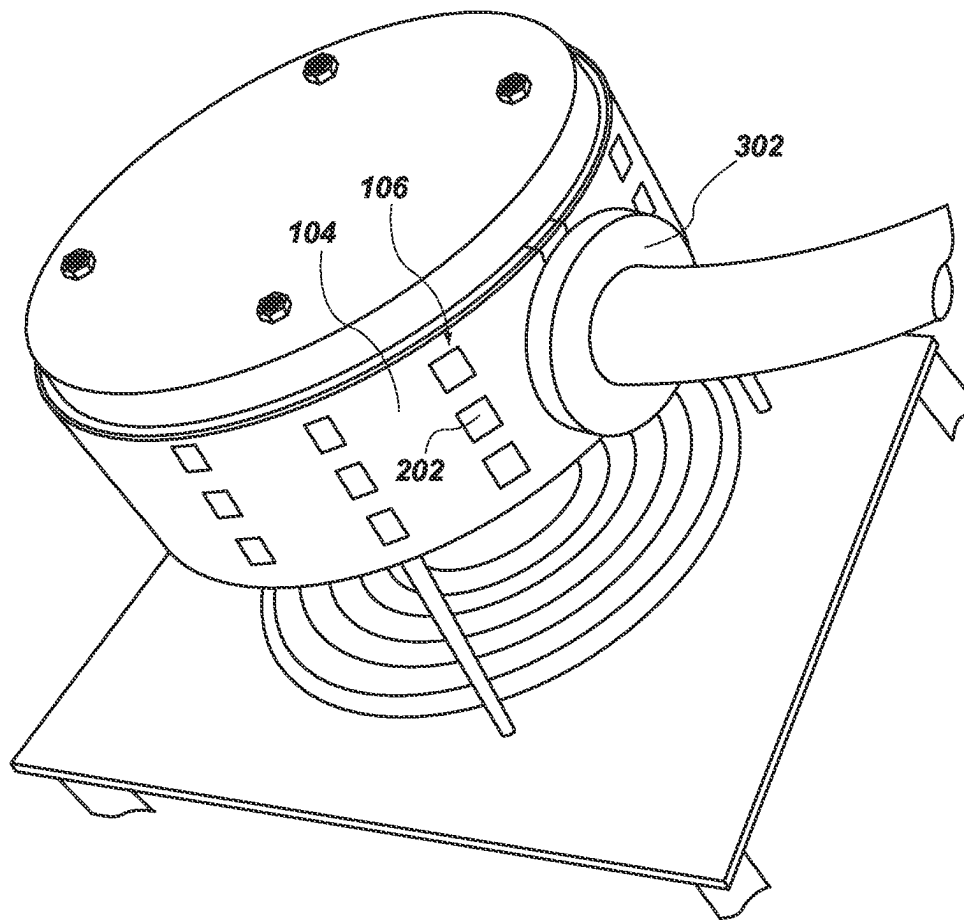
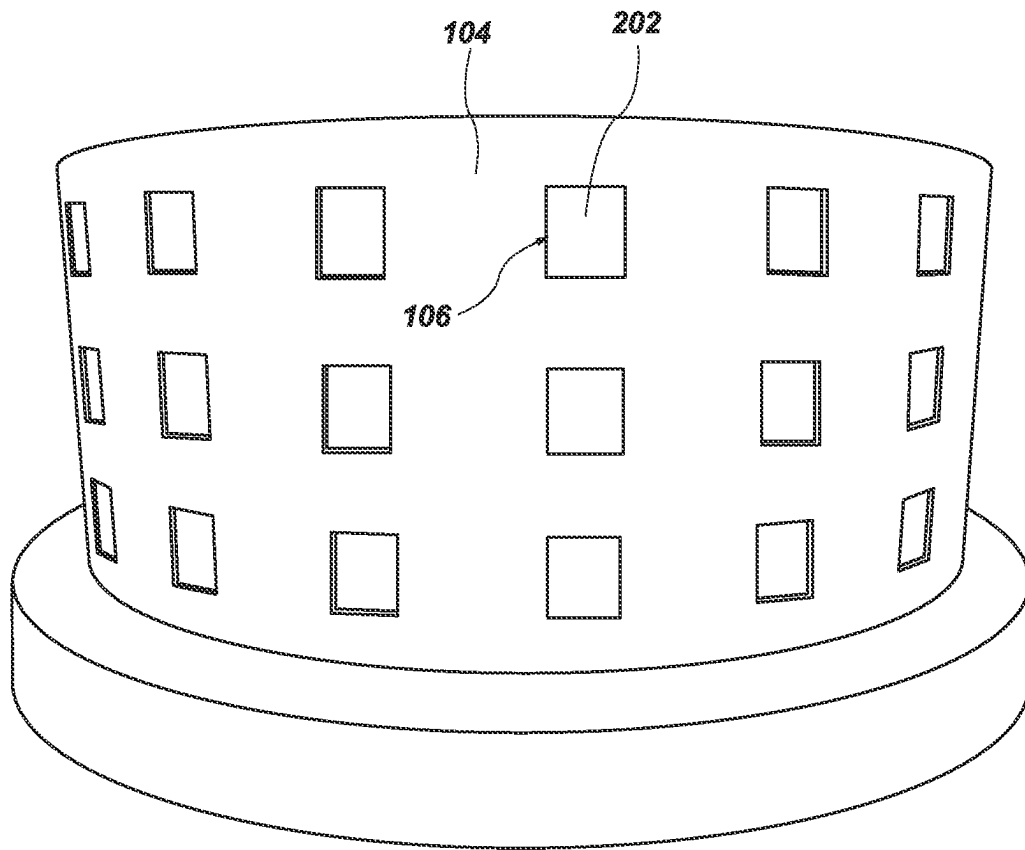


FIG. 1

**FIG. 2**

**FIG. 3**

**FIG. 4**

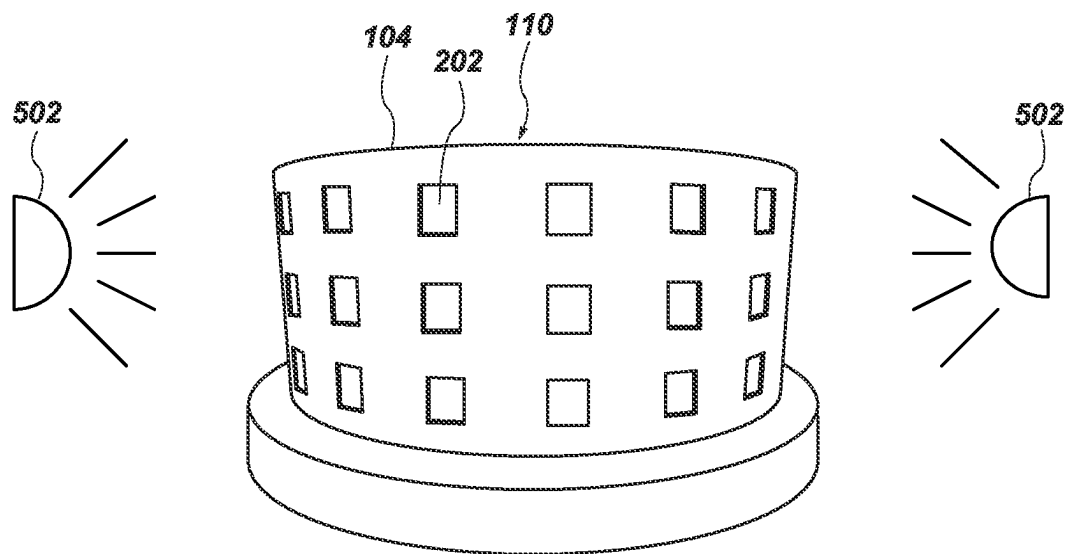


FIG. 5

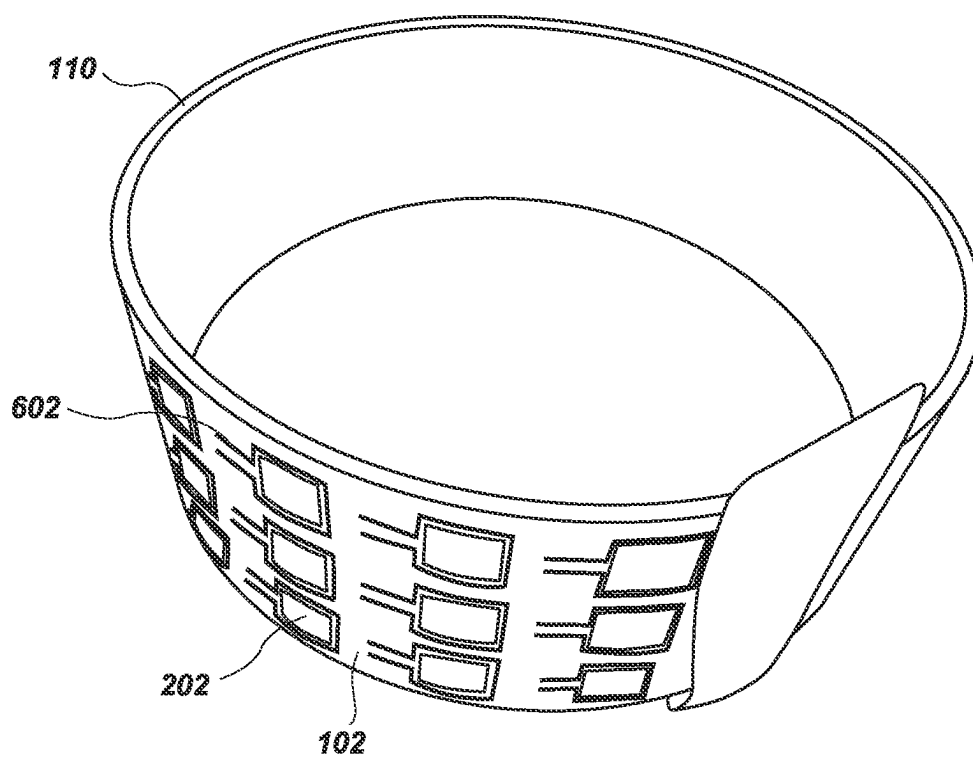


FIG. 6

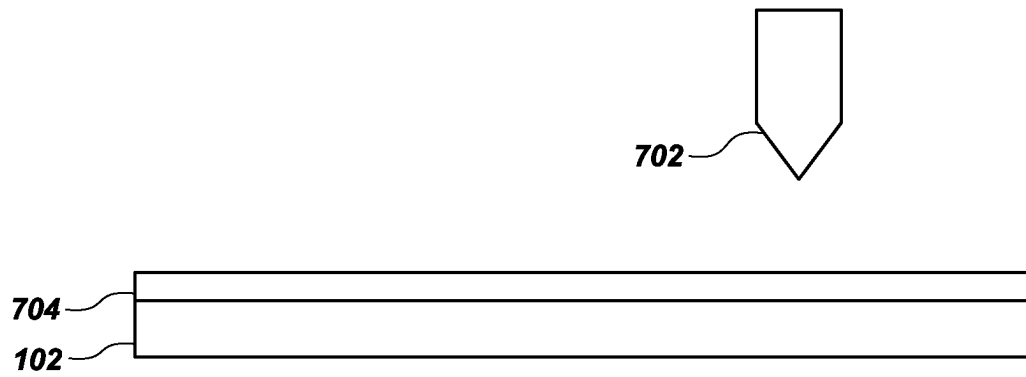


FIG. 7

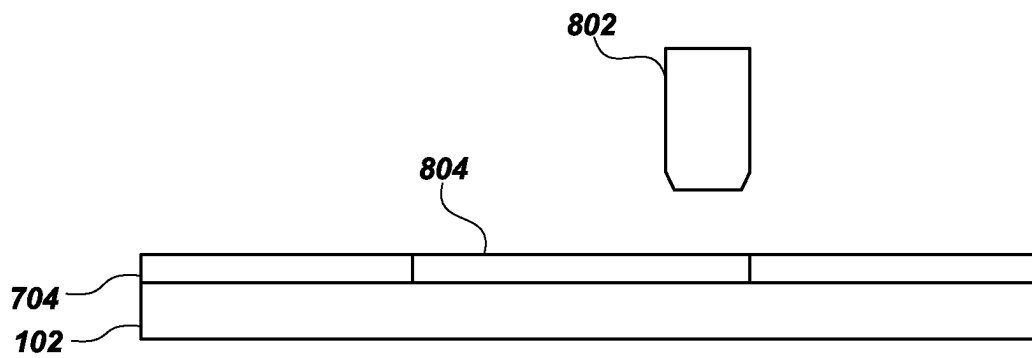


FIG. 8

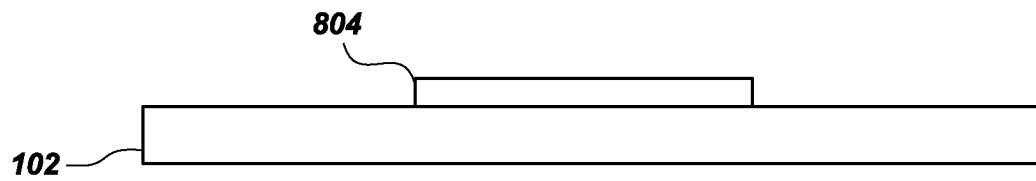


FIG. 9

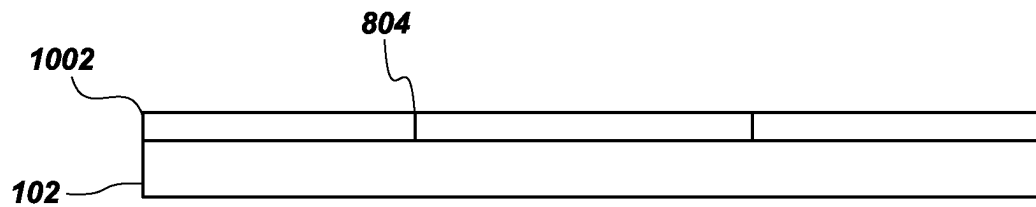


FIG. 10

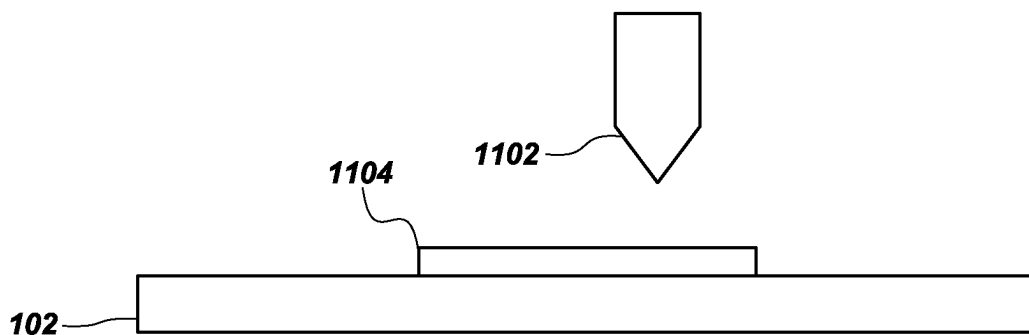


FIG. 11

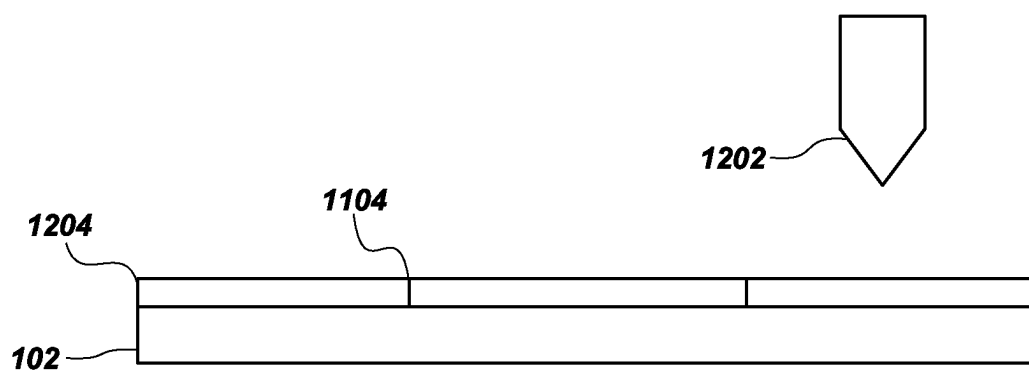


FIG. 12

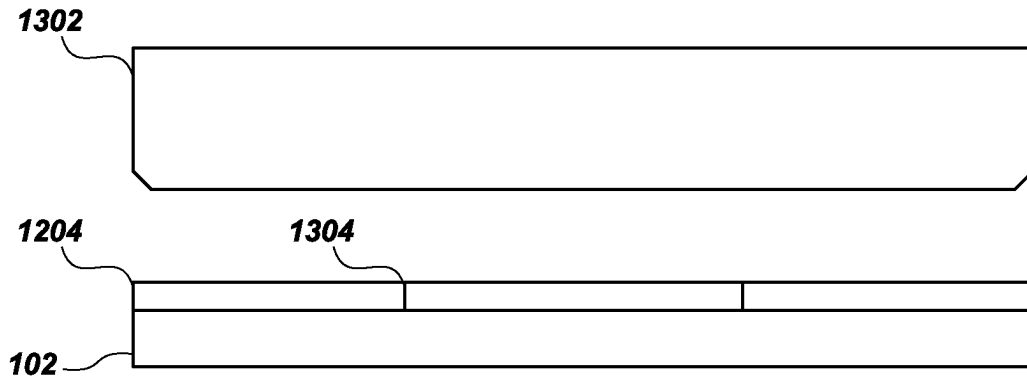


FIG. 13

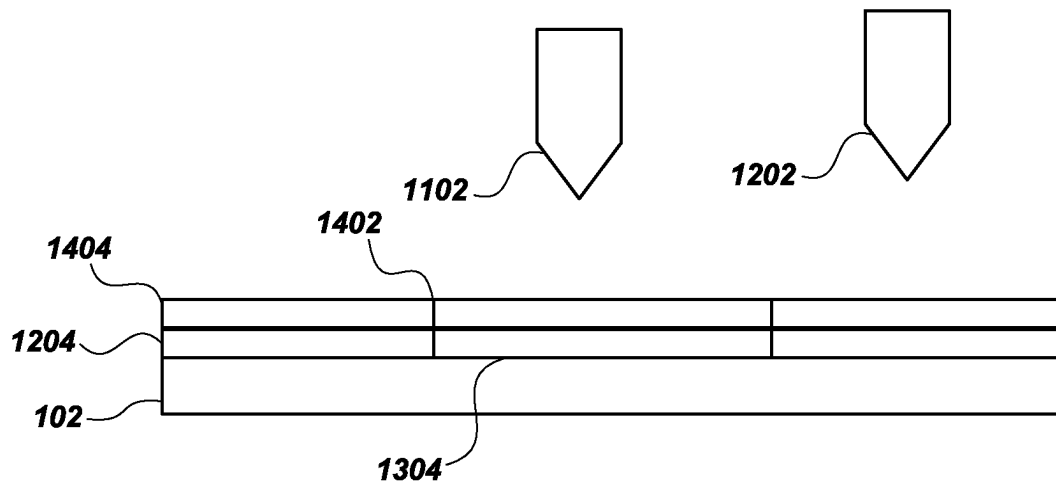


FIG. 14

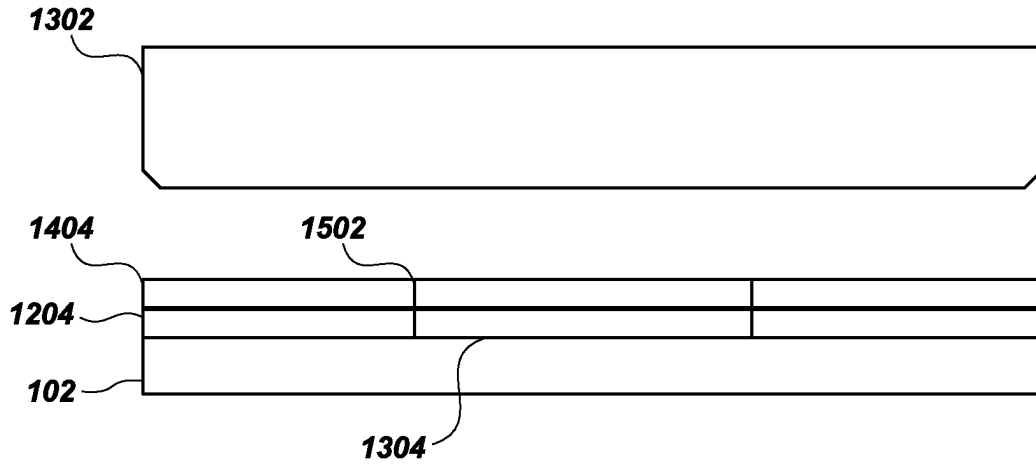


FIG. 15

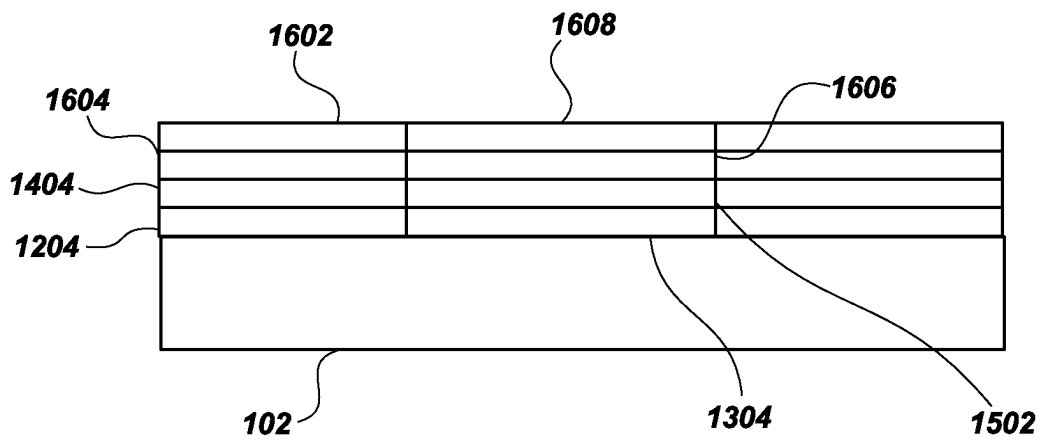


FIG. 16

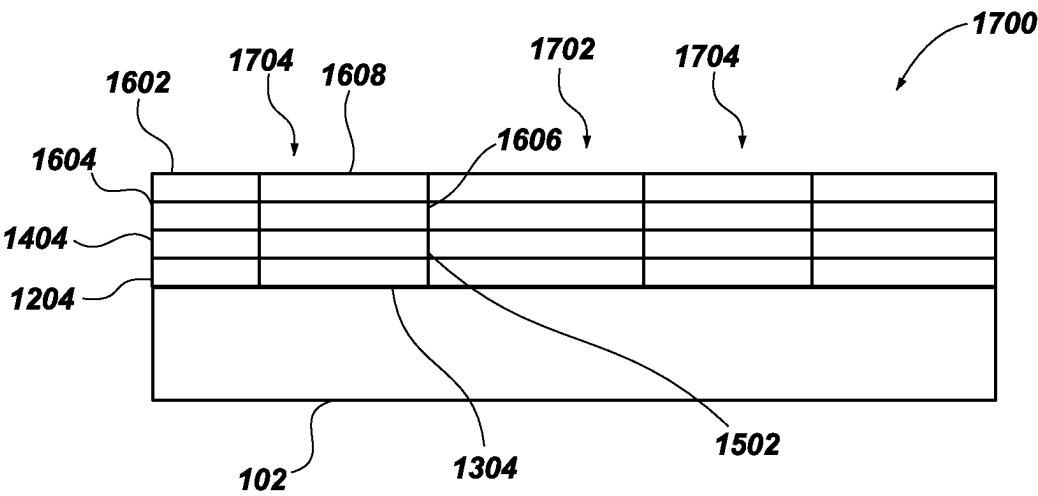


FIG. 17

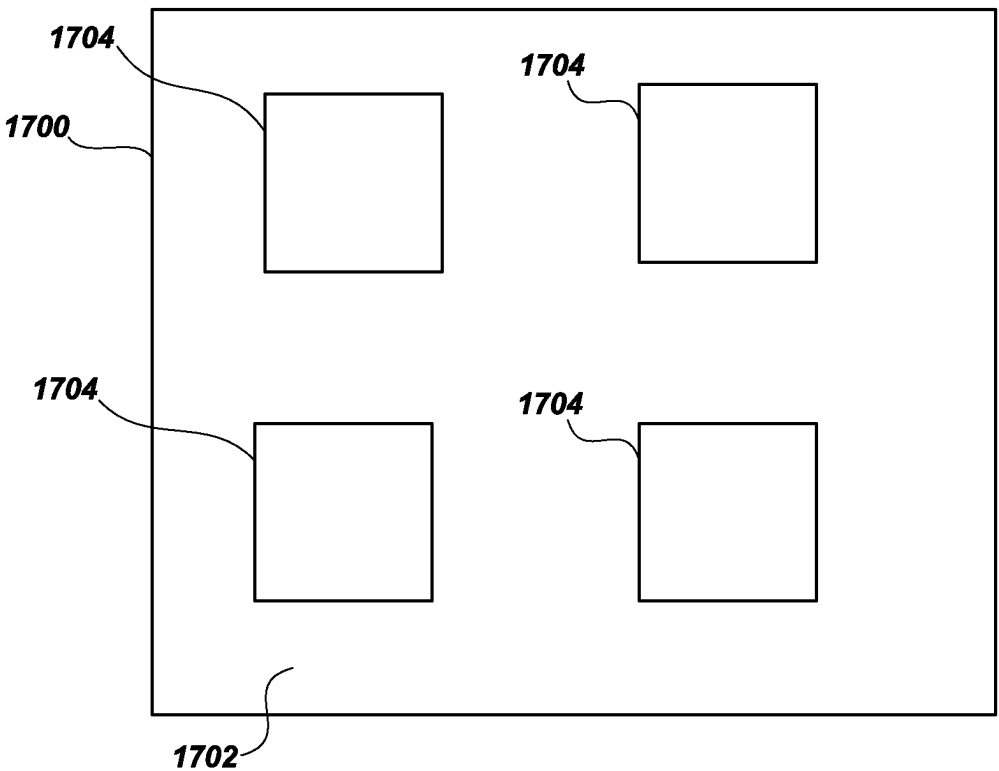


FIG. 18

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METHODS OF FORMING PIEZOELECTRIC MATERIALS, PIEZOELECTRIC DEVICES, AND ASSOCIATED TOOLING AND SYSTEMS

TECHNICAL FIELD

Embodiments of the present disclosure generally relate to methods of forming piezoelectric materials. In particular, embodiments of the present disclosure relate to methods of forming piezoelectric materials, piezoelectric devices, and associated tooling and systems.

BACKGROUND

Piezoelectric materials are materials configured to convert mechanical energy to electrical energy or electrical energy to mechanical energy through the piezoelectric effect. Piezoelectric devices may include sensing devices and/or actuating devices. For example, a piezoelectric sensing device may receive movement, such as vibrations or sound waves and convert the movement to an electrical signal that may be interpreted by an associated processor. A piezoelectric actuator may receive an electrical signal and move based on the electrical signal.

Piezoelectric devices may be used in imaging systems, such as ultrasound systems. For example, a piezoelectric device may be configured to generate the ultrasonic sound waves that travel through the item being imaged (i.e., person, device, formation). Another piezoelectric device may be configured to receive the ultrasonic sound waves and convert the waves to an electrical signal. The imaging system may then generate an image of an internal portion of the item being imaged from the electrical signal. Imaging systems utilizing piezoelectric devices may be used to image earth formations, such as for drilling operations (i.e., oil drilling, water drilling, geothermal drilling, etc.).

BRIEF SUMMARY

Embodiments of the present disclosure may include a method of forming a piezoelectric device. The method may include depositing a sol-gel film over a substrate. The method may further include curing the sol-gel film by impinging light onto an exposed surface of the sol-gel film to form a piezoelectric ceramic element.

Another embodiment of the present disclosure may include a piezoelectric composite material. The material may include at least two piezoelectric ceramic pillars over a substrate, the at least two piezoelectric pillars including at least one layer. The at least one layer having a gradient density, such that a first portion of the at least one layer proximate the substrate has a density lower than a second portion that is located a greater distance from the substrate than the first portion. The material may further include a resin separating the at least two piezoelectric ceramic pillars.

Another embodiment of the present disclosure may include a tool for forming a piezoelectric composite material. The tool may include a substrate support configured to secure a substrate. The tool may further include a first deposition device configured to deposit a sol-gel solution on the substrate. The tool may also include a second deposition device configured to deposit a resin on the substrate. The tool may further include a light configured to cure the sol-gel solution and the resin.

BRIEF DESCRIPTION OF THE DRAWINGS

While the specification concludes with claims particularly pointing out and distinctly claiming embodiments of the

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present disclosure, the advantages of embodiments of the disclosure may be more readily ascertained from the following description of embodiments of the disclosure when read in conjunction with the accompanying drawings in which:

FIGS. 1 through 6 illustrate tooling and process steps of forming a piezoelectric material in accordance with one or more embodiments of the present disclosure;

FIGS. 7 through 10 illustrate process steps for forming a piezoelectric material in accordance with one or more embodiments of the present disclosure;

FIGS. 11 through 15 illustrate process steps for forming a piezoelectric material in accordance with one or more embodiments of the present disclosure;

FIG. 16 illustrates a cross-sectional view of an embodiment of a piezoelectric device formed in accordance with one or more of the processes described in the present disclosure;

FIG. 17 illustrates a cross-sectional view of an embodiment of a piezoelectric composite material including the piezoelectric device of FIG. 16; and

FIG. 18 illustrates a top view of an embodiment of the piezoelectric composite material of FIG. 17.

DETAILED DESCRIPTION

The illustrations presented herein are not meant to be actual views of any particular earth-boring system or component thereof, but are merely idealized representations employed to describe illustrative embodiments. The drawings are not necessarily to scale.

As used herein, the term “substantially” in reference to a given parameter means and includes to a degree that one skilled in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances. For example, a parameter that is substantially met may be at least about 90% met, at least about 95% met, at least about 99% met, or even at least about 100% met.

As used herein, relational terms, such as “first,” “second,” “top,” “bottom,” etc., are generally used for clarity and convenience in understanding the disclosure and accompanying drawings and do not connote or depend on any specific preference, orientation, or order, except where the context clearly indicates otherwise.

As used herein, the term “and/or” means and includes any and all combinations of one or more of the associated listed items.

As used herein, the terms “vertical” and “lateral” refer to the orientations as depicted in the figures.

Piezoelectric materials may include ceramic materials, such as lead zirconate titanate (PZT, $\text{Pb}[\text{Zr}_x\text{Ti}_{1-x}]\text{O}_3$ with $0 \leq x \leq 1$), potassium niobate (KNbO_3), sodium tungstate (Na_2WO_3), zinc oxide (ZnO), etc., crystalline materials, such as quartz, langasite ($\text{La}_3\text{Ga}_5\text{SiO}_{14}$), gallium orthophosphate (GaPO_4), lithium niobate (LiNbO_3), lithium tantalate (LiTaO_3), lead titanate (PbTiO_3), and/or composite materials including piezoelectric ceramic materials or piezoelectric crystalline materials and a polymer, etc. When forming ceramic piezoelectric materials a solution may be formed that may then need to be cured to form the ceramic piezoelectric material. Common curing techniques involve heating (e.g., pyrolysis and sintering) the solution to form the ceramic piezoelectric material. The high temperatures involved in heating the solution may cause lead within the solution and/or ceramic material to volatilize. This is commonly referred to as “lead loss” in the art. Lead loss can

reduce the performance of the resulting piezoelectric device by causing a stoichiometric imbalance in the piezoelectric device. The stoichiometric imbalance may reduce the electro-acoustic performance thereof.

The methods described herein may enable piezoelectric devices to be formed from ceramic piezoelectric materials in a manner that significantly reduces the "lead loss" of the associated piezoelectric device. Furthermore, the methods described herein may further increase the efficiency of the piezoelectric device formation, such as by reducing the amount of waste material and/or reducing tooling requirements. This may increase the efficiency of forming high surface area composite structures.

FIG. 1 illustrates an apparatus **100** for forming multiple piezoelectric elements. The apparatus **100** may include a drum **110** supporting a substrate **102**, on which the piezoelectric elements may be formed. A mask **104** may be secured over the substrate **102** with one or more clamping mechanisms **108**. The mask **104** may include one or more apertures **106** extending through the mask **104** exposing the substrate **102**.

The drum **110** may be a cylindrical support configured to support the substrate **102** on an exterior surface of the drum **110**. The drum **110** may be configured to rotate the substrate **102** and mask **104**, such that a deposition apparatus may be used to coat the exposed portions of the substrate **102** with a solution. Rotating the drum **110** may enable the deposition apparatus to remain substantially stationary while coating the exposed portions of the substrate **102** about the entire outer surface of the drum **110**.

The solution may be deposited on the exposed portions of the substrate **102** through a chemical solution deposition process, such as a sol-gel process. In a sol-gel deposition process a solution including the elements of the desired piezoelectric material may be formed. For example, a solution including lead, zirconium, and titanium (e.g., a solution of zirconium butoxide, titanium butoxide, and lead acetate trihydrate) may be used to create a sol-gel solution for forming a PZT material. In some embodiments, the sol-gel solution may further include a powder, such as a PZT powder (e.g., bismuth titanate, modified bismuth titanate, bismuth scandium lead titanate, lithium niobate and other high temperature piezoelectric powder materials), mixed into the solution. The powder may be fully reacted and dispersed in a sol-gel solution. The powder may reduce the amount of volume lost through pyrolysis when dipping, drying and curing multiple layers. Reducing the volume loss may improve the stability of the sol-gel solution, which may substantially reduce failures of the resulting devices, such as from separation from the substrate, cracks, etc., caused by thermal expansion during the drying and curing processes.

The sol-gel solution may then be deposited onto the mask **104** and substrate **102** through the deposition apparatus, through a process, such as dipping, spraying, sputtering, etc. The deposition apparatus may be configured to create a layer (e.g., film or coating) of the sol-gel solution that is substantially uniform across the surface of the mask **104** and the exposed portions of the substrate **102**, as illustrated in FIG. 2.

After the mask **104** and exposed portions of the substrate **102** are coated with the sol-gel solution, the drum **110** may be removed and allowed to dry as illustrated in FIG. 2. As the sol-gel solution dries, the sol-gel solution may form a semi-solid or gel-like material, such that the sol-gel solution may remain flexible while remaining in substantially the same position on the mask **104** and substrate **102**.

The drying process may remove water and methanol from the sol-gel solution. The removal of the water and methanol may lead to the formation of lead oxide. In some embodiments, the drying process may occur at an elevated temperature, such as a temperature of between about 100° C. and about 600° C. The elevated temperature may increase the amounts of water and methanol removed and may similarly increase the formation of lead oxide. Furthermore, the increased temperature may cause species within the sol-gel to decompose causing crystallization to occur within the sol-gel.

In some embodiments, the elevated temperatures may be staged. For example, a first temperature of between about 150° C. and about 250° C., such as about 200° C., may be used to remove water and methanol and begin forming the lead oxide. A second temperature of between about 270° C. and about 350° C. may cause other species within the sol-gel to decompose. A final temperature of between about 450° C. and about 560° C. may cause pyrochlore and/or perovskite crystallization to occur within the sol-gel material. The crystallization may increase the rigidity of the semi-solid sol-gel material.

The portions of the sol-gel solution in the apertures **106** of the mask **104** may form an element **202** on the substrate **102** that was exposed by the apertures **106** of the mask **104**. The element **202** may be a semi-solid element of a substantially uniform thickness supported on the perimeter edges by the walls of the respective apertures **106** of the mask **104**.

Each element **202** may have a shape substantially the same as the respective aperture **106**. For example, if the aperture **106** has a square shape as illustrated in FIGS. 1-5, the respective element **202** may also have a square shape. In other embodiments, the apertures **106** and respective elements **202** may have other shapes, such as triangular shapes, rectangular shapes, circular shapes, etc.

After the sol-gel solution is dried, the mask **104** and exposed portions of the substrate **102** that are coated with the sol-gel solution may be cooled as illustrated in FIG. 3. A cooling apparatus **302** may be configured to cool the sol-gel solution. The cooling apparatus **302** may include one or more of a ventilation system, a refrigerant system, a cold plate, etc. Cooling the sol-gel solution may further solidify the sol-gel solution both on the mask **104** and in the elements **202** within the apertures **106** of the mask **104**.

After the sol-gel solution is dried and cooled, a second layer of sol-gel solution may be deposited over the dried and cooled sol-gel solution. The second layer of sol-gel solution may go through substantially the same processing steps as the first sol-gel solution layer going through both a drying and a cooling step. The same processing steps may be repeated until the element **202** reaches a desired thickness. For example, the processing steps may be repeated until the element **202** reaches a thickness of between about 9 μm and about 500 μm , such as between about 50 μm and about 200 μm .

The thickness of the element **202** may substantially correlate to a final thickness of the resulting ceramic material in the piezoelectric device. The thickness of the ceramic material in the piezoelectric device may affect the performance of the piezoelectric device. For example, a thinner ceramic material may be more responsive to electrical signals and/or movement, whereas a thicker ceramic material may be more robust and able to withstand greater amounts of movement and/or larger electrical signals. For audio devices, such as ultrasound imaging devices a thinner ceramic material may

enable the use of higher frequencies (i.e., the thinner ceramic material may be able to detect and/or produce higher frequencies).

Once the desired thickness is reached, the sol-gel solution may be cured as illustrated in FIG. 5. Curing the sol-gel solution may cause the sol-gel solution to calcine and solidify into a substantially rigid ceramic material, such as PZT. The sol-gel solution may be cured by impinging a light 502 on the sol-gel solution. The light 502 may be in the ultraviolet or visible spectrums, such as light having wavelengths of between about 100 nanometers (nm) and about 800 nm, such as between about 200 nm and about 600 nm, or between about 300 nm and about 500 nm. The light 502 may cause the sol-gel solution to calcine without heating the sol-gel solution to high temperatures (e.g., temperatures of greater than about 700° C.). As described above, reducing the curing temperature may reduce the amounts of lead loss experienced by the ceramic material during the curing process.

In some embodiments, the drum 110 with the substrate 102, mask 104, and elements 202 may be placed in an enclosed area, such as a box surrounded by lights configured to emit light in the ultraviolet or visible spectrum, such that the light may impinge on the sol-gel solution deposited over the substrate 102 and mask 104.

The light 502 may impinge on the surface of the sol-gel solution for a time period between about 1 s and about 1 hr, such as between about 1 s and about 120 s. The time period may be determined based on parameters such as material thickness, capacitor bank voltage, pulse length, pulse repetition rate, pulse number, average power exposure, etc.

After the sol-gel solution is cured forming the ceramic material, the process may be repeated to form multiple layers of ceramic material until a desired thickness is reached. After the multiple layers of ceramic material are created, the mask 104 may be removed as illustrated in FIG. 6. FIG. 6 illustrates the substrate 102 over the drum 110 with the elements 202 formed thereon. The cured elements 202 may have a thickness of between about 5 μm and about 40 μm, such as between about 9 μm and about 12 μm.

The substrate 102 may be formed from a substantially heat resistant and flexible material, such as a metal (e.g., stainless steel) or a polymer. The substrate 102 may include contacts 602 formed on the surface of the substrate 102. For example, the contacts 602 may be formed from a conductive material, such as copper or gold, in a process step before forming the elements 202. The contacts 602 may be formed in a process such as sputtering. After the contacts 602 are formed the elements 202 may be formed over the contacts, in any of the processes described herein. In some embodiments, the substrate 102 may be a conductive material, such that the substrate 102 may form an electrode for the resulting piezoelectric material. After the elements 202 are formed a top contact may be formed over the elements 202, such as through an additional sputtering process.

In some embodiments, the piezoelectric ceramic material may be formed without using a mask 104. FIG. 7 through FIG. 10 illustrate an example of a method of forming the piezoelectric ceramic material over the substrate 102 without a mask 104. First, a sol-gel layer 704 (e.g., film, coating, etc.) may be formed over the substrate 102. Similar to the process described above, the sol-gel layer 704 may be formed through a deposition device 702, such as a spraying device, a sputtering device, a dipping device (e.g., dipping the substrate 102 into a sol-gel solution), etc.

In some embodiments the sol-gel layer 704 may be formed of multiple layers of sol-gel solution. For example,

a first layer may be applied and allowed to dry before a second layer may be applied over the first layer. The application may be repeated until the sol-gel layer 704 reaches the desired thickness. As described above, the drying process may occur at an elevated temperature. In other embodiments, the drying process may occur at a temperature close to a typical room temperature.

After the sol-gel layer 704 is formed a light 802 may be used to selectively cure a portion of the sol-gel layer 704 to form a ceramic element 804 as illustrated in FIG. 8. The light 802 may be a focused light, such as a laser, configured to control the area upon which the light 802 impinges. Focusing the light may enable the light 802 to be used to define the shape of the ceramic element 804 by selectively curing the sol-gel layer 704 in the area where the ceramic element 804 is desired and leaving the sol-gel layer 704 outside the ceramic element 804 substantially uncured.

The light 802 may be in the ultraviolet or visible spectrums, such as light having wavelengths of between about 100 nm and about 800 nm, such as between about 200 nm and about 600 nm, or between about 300 nm and about 500 nm. The light 802 may cause the sol-gel solution in the selected area to calcinate. For example, the focused light 802 may excite the sol-gel solution raising the local temperature to the calcination temperature without raising the temperature surrounding the sol-gel solution. Raising the local temperature may cause the area of the sol-gel solution in direct contact with the focused light 802 to calcinate, sintering the sol-gel solution into the ceramic element 804 without heating the area around the sol-gel solution.

In some embodiments, the light 802 may be used to execute all of the drying and curing processes for forming the ceramic element 804. For example, as described above, different elevated temperatures may produce different effects in the curing process of the sol-gel layer 704. The light 802 may be used to create local elevated temperatures for similar effects, such as removing water and methanol and beginning to form the lead oxide, causing other species within the sol-gel to decompose, and causing pyrochlore and/or perovskite crystallization to occur within the sol-gel material. Curing and/or drying the ceramic element 804 with the light may create a density gradient within the ceramic element 804, such that the density of the ceramic element 804 may be different at different locations in the ceramic element 804. For example, the density of the ceramic element 804 proximate the substrate 102 may be less than the density of the ceramic element proximate an upper surface of the ceramic element 804 that is on an opposite side of the ceramic element from the substrate 102.

As described above, reducing the heat in the areas around the sol-gel material may substantially reduce the lead loss experienced by the sol-gel solution during the curing process. Furthermore, reducing the heat in the areas around the sol-gel solution may enable materials with a lower tolerance for high temperatures to be used in conjunction with the piezoelectric ceramic materials, such as for substrates and/or contacts on the substrates.

In some embodiments, the ceramic element 804 may be formed from multiple sol-gel layers 704. For example, a first sol-gel layer 704 may be deposited and cured as described above. After the first sol-gel layer 704 is formed and selectively cured, a second sol-gel layer 704 may be formed over the first sol-gel layer 704. The second sol-gel layer 704 may also be selectively cured by the light 802 as described above. The process may be repeated until the ceramic element 804 reaches a desired thickness.

In some cases, each layer may only be partially cured before the ceramic element **804** reaches the desired thickness. For example, the light **802** may induce one or more of the intermediate drying/curing effects described above, such as removing water and methanol and beginning to form the lead oxide, causing other species within the sol-gel to decompose, and/or causing pyrochlore and/or perovskite crystallization to occur within the sol-gel material, without causing the respective sol-gel layer **704** to calcinate and fully sinter into the ceramic element **804**. In other cases, each layer may be fully cured or sintered before the next layer is applied.

After the ceramic element **804** is cured by the light **802**, the remaining sol-gel layer **704** may be removed as illustrated in FIG. 9. For example, the remaining sol-gel layer **704** may be removed by washing the substrate **102**, such as in a water or solvent bath. The washing process may be configured to remove the uncured sol-gel layer **704** while leaving the cured ceramic element **804** substantially unaffected. Thus, after the washing process, the ceramic element **804** may remain over the substrate **102**.

After the remaining sol-gel layer **704** is removed from the area surrounding the ceramic element **804**, a resin **1002** may be deposited around the ceramic element **804**. The resin **1002** may be an epoxy resin or a UV curable resin. In some embodiments, as further described below, the substrate **102** may include an array of multiple ceramic elements **804** separated by the resin **1002** to form a piezoelectric composite material. Forming the piezoelectric composite material in the manner described above, may enable smaller scale composite structures. For example, the composite structures may have a kerf width (e.g., distance between ceramic elements **804**) of between about 15 μm and about 380 μm and a pitch (e.g., distance from a first side of a ceramic element **804** to a first side of an adjacent ceramic element **804**) of between about 70 μm and about 850 μm .

FIG. 11 through FIG. 15 illustrate another example of a method of forming the piezoelectric ceramic material over the substrate **102** without a mask **104**. A first sol-gel layer **1104** may be formed over the substrate **102**. The first sol-gel layer **1104** may be formed by a sol-gel deposition device **1102** configured to selectively deposit the first sol-gel layer **1104**. For example, the sol-gel deposition device **1102** may be a deposition head similar to a printing head of a 3-D printer configured to deposit the first sol-gel layer **1104** in a specific pattern over the substrate **102**.

The sol-gel deposition device **1102** may move over the substrate **102** or the substrate **102** may move relative to the sol-gel deposition device **1102**, such that the sol-gel deposition device **1102** may travel in a grid pattern relative to the substrate **102** only depositing the first sol-gel layer **1104** in the regions where the piezoelectric ceramic material is desired and leaving the other portions of the substrate **102** substantially free of the sol-gel material.

As illustrated in FIG. 12, a resin deposition device **1202** may be configured to deposit a first resin layer **1204** over the substrate **102** in the regions that are substantially free of the first sol-gel layer **1104**. The first resin layer **1204** may include a resin material, such as an epoxy resin or a UV curable resin. For example, the first resin layer **1204** may be formed from a resin configured to cure under lights having wavelengths in the ultraviolet spectrum.

In some embodiments, the resin deposition device **1202** may be configured to operate simultaneously with the sol-gel deposition device **1102**. For example, a deposition head may include both the sol-gel deposition device **1102** and the resin deposition device **1202**, such that as the deposition

head moves relative to the substrate **102** the deposition head may selectively deposit sol-gel from the sol-gel deposition device **1102** or resin from the resin deposition device **1202** based on a location on the substrate **102**.

In other embodiments, the resin deposition device **1202** may be configured to operate separately from the sol-gel deposition device **1102**. For example, the sol-gel deposition device **1102** may move independently from the resin deposition device **1202** (e.g., on a separate deposition head). In some cases, the sol-gel deposition device **1102** may deposit the first sol-gel layer **1104** before the resin deposition device **1202** deposits the first resin layer **1204**. In other cases, the sol-gel deposition device **1102** may deposit the first sol-gel layer **1104** at substantially the same time as the resin deposition device **1202** deposits the first resin layer **1204**, however, the sol-gel deposition device **1102** may move independently from the resin deposition device **1202**, such as along different paths, or in different areas.

After the first resin layer **1204** and the first sol-gel layer **1104** are deposited over the substrate **102**, a light **1302** may be used to cure the first resin layer **1204** and the first sol-gel layer **1104** as illustrated in FIG. 13. As described above, the light **1302** may cure the first sol-gel layer **1104** to form a first ceramic layer **1304** by removing water and methanol, beginning to form the lead oxide, causing other species within the sol-gel to decompose, causing pyrochlore and/or perovskite crystallization to occur within the sol-gel material, and/or causing the first sol-gel layer **1104** to calcinate and fully sinter into the first ceramic layer **1304**.

As described above, the first resin layer **1204** may be a resin configured to cure under ultraviolet light. Therefore, the light **1302** may be configured to produce light having wavelengths within the ultraviolet spectrum, such that the light **1302** may be configured to cure both the first sol-gel layer **1104** and the first resin layer **1204** substantially simultaneously. For example, the light **1302** may produce light having wavelengths between about 100 nm and about 400 nm, such as wavelengths between about 300 nm and about 400 nm.

After the first resin layer **1204** and the first ceramic layer **1304** are cured, the sol-gel deposition device **1102** and the resin deposition device **1202** may deposit a second sol-gel layer **1402** and a second resin layer **1404** over the first resin layer **1204** and the first ceramic layer **1304**, as illustrated in FIG. 14. The second sol-gel layer **1402** and the second resin layer **1404** may be deposited in substantially the same manner as the first sol-gel layer **1104** and the first resin layer **1204**, as described above.

After the second sol-gel layer **1402** and the second resin layer **1404** are deposited over the first resin layer **1204** and first ceramic layer **1304**, the light **1302** may be used to cure the second sol-gel layer **1402** and the second resin layer **1404** as illustrated in FIG. 15. As described above, the light **1302** may cure the second sol-gel layer **1402** to form a second ceramic layer **1502** by forming lead oxide, causing other species within the sol-gel to decompose, causing pyrochlore and/or perovskite crystallization to occur within the sol-gel material, and/or causing the second sol-gel layer **1402** to calcinate and fully sinter into the second ceramic layer **1502**.

As described above, the second resin layer **1404** may be a resin configured to cure under ultraviolet light. Therefore, the light **1302** may be configured to produce light having wavelengths within the ultraviolet spectrum, such that the light **1302** may be configured to cure both the second resin layer **1404** and the second sol-gel layer **1402** substantially simultaneously.

The processes of depositing layers the sol-gel material and the resin and curing the sol-gel and resin layers may be repeated until a device having the desired thickness is formed as illustrated in FIG. 16. The final device may include multiple layers of resin **1204**, **1404**, **1604**, **1602** and multiple layers of ceramic, **1304**, **1502**, **1606**, **1608**.

Curing each of the layers independently with the light **1302** as described above, may result in the ceramic layers **1304**, **1502**, **1606**, **1608** having substantially the same density. Each of the ceramic layers **1304**, **1502**, **1606**, **1608** may have a density gradient as described above, such that a portion each of the ceramic layers **1304**, **1502**, **1606**, **1608** nearer (e.g., proximate) the substrate **102** may have a lower density than the portions of each of the ceramic layers **1304**, **1502**, **1606**, **1608** that are further from the substrate **102**. The light **1302** may impinge on the top of each layer, such that intensity of the light may reduce as the light travels through each ceramic layer. The reduction in the intensity of the light may result in a reduction in the density of the lower portions of the ceramic layers **1304**, **1502**, **1606**, **1608**.

The ceramic layers **1304**, **1502**, **1606**, **1608** may combine to form one or more pillars **1704** over the substrate **102**, as illustrated in FIG. 17. The pillars **1704** may be separated by resin **1702** (i.e., the layers of resin **1204**, **1404**, **1604**, **1602**). A pattern of pillars **1704** may form an array as illustrated in FIG. 18. The array of pillars **1704** separated by resin **1702** may form a piezoelectric composite material **1700**. The pillars **1704** may be electrically coupled to one another through contacts formed on the substrate **102**, such as the contacts **602** described above. The contacts may enable the pillars **1704** to generate an electrical signal from the piezoelectric composite material **1700** and/or respond to an electrical signal applied to the piezoelectric composite material **1700**. In some embodiments, as described above, the substrate **102** may be formed from a conductive material, such that the substrate **102** may form an electrical connection between the pillars **1704** and act as an electrode for the piezoelectric composite material **1700**.

Embodiments of the present disclosure may enable a piezoelectric material to be formed without high temperature sintering. Forming the piezoelectric material without high temperature sintering may reduce the amount of lead loss during the curing process, which may result in higher performance materials. Furthermore, reducing the processing temperatures may enable the use of materials having a lower tolerance for heat, such as for substrates, contacts, resins, etc.

The embodiments of the disclosure described above and illustrated in the accompanying drawing figures do not limit the scope of the invention, since these embodiments are merely examples of embodiments of the invention, which is defined by the appended claims and their legal equivalents. Any equivalent embodiments are intended to be within the scope of this disclosure. Indeed, various modifications of the present disclosure, in addition to those shown and described herein, such as alternative useful combinations of the elements described, may become apparent to those skilled in the art from the description. Such modifications and embodiments are also intended to fall within the scope of the appended claims and their legal equivalents.

What is claimed is:

1. A method of forming a piezoelectric device comprising: applying a substrate over a cylindrical surface of a body of a drum;

applying a mask over the substrate around the cylindrical surface of the body of the drum;

depositing a sol-gel film including a piezoelectric material over the substrate through openings defined in the mask;

curing the sol-gel film by impinging light onto an exposed surface of the sol-gel film to form a piezoelectric ceramic element.

2. The method of claim 1, wherein the light has a wavelength between about 800 nanometers (nm) and about 100 nm.

3. The method of claim 1, wherein the sol-gel film is deposited over the mask positioned over the substrate, the openings configured to enable the sol-gel film to pass through the openings to the substrate.

4. The method of claim 1, wherein the light is selectively impinged over a portion of the sol-gel film, such that the portion of the sol-gel film where the light is selectively impinged forms the piezoelectric ceramic element.

5. The method of claim 1, wherein the sol-gel film is selectively deposited over the substrate, such that a portion of the substrate is substantially free from the sol-gel film.

6. The method of claim 5, further comprising depositing a resin over the portion of the substrate that is substantially free from the sol-gel film.

7. The method of claim 6, wherein the resin is an ultraviolet curable resin.

8. The method of claim 1, further comprising depositing a second sol-gel film over the sol-gel film to form a second layer.

9. The method of claim 8, wherein the second sol-gel film is deposited after curing the sol-gel film.

10. The method of claim 1, wherein the sol-gel film comprises a solution of zirconium butoxide, titanium isopropoxide, and lead acetate trihydrate.

11. The method of claim 10, wherein the sol-gel film comprises a PZT powder.

12. A tool for forming a piezoelectric composite material comprising:

a substantially cylindrical substrate support configured to secure a substrate;

a mask configured to clamp the substrate to the cylindrical substrate support, the mask defining openings configured to expose the substrate beneath the mask;

a first deposition device configured to deposit a sol-gel solution on the substrate through the openings defined in the mask;

a second deposition device configured to deposit a resin on the substrate; and

a light configured to cure the sol-gel solution and the resin.

13. The tool of claim 12, wherein the second deposition device is configured to deposit the resin at substantially a same time as the first deposition device deposits the sol-gel solution.

14. The tool of claim 12, wherein the first deposition device and the second deposition device are configured to move independent of one another.

15. The tool of claim 12, wherein the first deposition device and the second deposition device are coupled to a single deposition head.

16. The tool of claim 12, wherein the light is configured to output light in wavelengths between about 800 nanometers (nm) and about 100 nm.

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